NATIONAL ECONOMICS UNIVERSITY
COLLEGE OF TECHNOLOGY

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MID-TERM ASSIGNMENT

COURSE: Data Analysis with Python

TOPIC: CLUSTERING EMPLOYEES BY WORK BEHAVIOUR AND SATISFACTION – HR ANALYTICS CASE STUDY

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Abstract

This report applies unsupervised learning to the *HR Analytics Case Study* dataset, which describes about 4,410 employees of a company through demographic, job, survey and attendance (swipe-in/swipe-out) records.

The goal is to discover naturally occurring employee segments without using any pre-defined label. We integrate the relational files, clean and engineer features (including average daily working hours derived from attendance logs), and compare three clustering algorithms — K-Means, Hierarchical (Ward) clustering and DBSCAN — using the Silhouette score, the Elbow method, Davies–Bouldin and Calinski–Harabasz indices.

The resulting segments are profiled and cross-checked against the `Attrition` column (used only for external validation) to derive practical HR recommendations.

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1 Initial Data Inspection

Before data preprocessing, an initial inspection was conducted to understand the structure and quality of all five datasets. The inspection focused on the consistency of the files, the relationships between them, file encoding, data types, missing values, and duplicate records.

1.1 Dataset Overview

The HR Analytics dataset consists of five relational files: `general`, `employee`, `manager`, `in_time`, and `out_time`. Each file contains 4,410 records.

Table 1 summarizes the basic information of the five datasets.

Table 1: Overview of the five datasets

Criteria	General	Employee	Manager	In-time	Out-time
Records	4,410	4,410	4,410	4,410	4,410
Attributes	24	4	3	262	262
Missing values	28	83	0	109080	109080
Duplicate records	0	0	0	0	0

Across the five files, there are 555 attributes in total. However, the 22,050 records represent the sum of records across files, not the number of unique employees, since the same employees are represented in multiple relational files.

1.2 Column Consistency

The five files describe different information, so they cannot be concatenated; they must be merged later. Consistency was therefore checked in two ways. First, the numbers of rows and columns were compared (Table 1): every file has 4,410 rows, while the number of columns differs by file (24, 4, 3, 262 and 262). Second, the column names of `in_time` and `out_time`, which have the same structure (one column per calendar day), were compared. They contain the same 262 column names in the same order, so the two files can be processed in parallel when computing working hours. No column name other than `EmployeeID` is shared between the other files, so no naming conflict (e.g. `_x`, `_y` suffixes) will appear after merging.

1.3 Relationships Between Files

The common column of the five files is `EmployeeID`. In `in_time` and `out_time` this column is stored in an unnamed first column (`Unnamed: 0`) and is renamed to `EmployeeID`. Table 2 shows the key checks.

Table 2: Primary key / foreign key check on `EmployeeID`

File	Null keys	Duplicate keys	IDs not in General	General IDs missing
General	0	0	0	0
Employee	0	0	0	0
Manager	0	0	0	0
In-time	0	0	0	0
Out-time	0	0	0	0

`EmployeeID` is the **primary key** of the `general` dataset (unique and non-null) and the **foreign key** in the other four files, so the relationship is one-to-one. A test merge of `general`, `employee` and `manager` on `EmployeeID` returned 4,410 rows, i.e. no record was lost or duplicated.

1.4 File Encoding

The encoding of each CSV file was tested by decoding the raw bytes. All five files were successfully decoded as UTF-8 (no BOM, no decoding error), so no special encoding argument is needed when calling `pd.read_csv()` and no font or character errors are expected.

1.5 Data Types

The data types of each dataset were examined using the `info()` method.

Table 3: Distribution of data types across datasets

Dataset	int64	float64	str	Total attributes
General	14	2	8	24
Employee	1	3	0	4
Manager	3	0	0	3
In-time	1	12	249	262
Out-time	1	12	249	262
Total	**20**	**29**	**506**	**555**

The `general` dataset contains demographic and job-related information. The `employee` dataset provides employee satisfaction and work-life balance information, while the `manager` dataset contains job involvement and performance ratings.

The `in_time` and `out_time` datasets each contain 262 attributes, including 249 string attributes, 12 floating-point attributes, and 1 integer attribute. These two files contain attendance-related records and will be examined further during feature engineering.

1.6 Missing Values

Missing values were identified in the `general` and `employee` datasets.

1.6.1 General Dataset

The `general` dataset contains 28 missing values across two attributes.

Table 4: Missing values in the General dataset

Attribute	Missing values	Percentage
NumCompaniesWorked	19	0.43%
TotalWorkingYears	9	0.20%
Total	**28**	**0.63%**

1.6.2 Employee Dataset

The `employee` dataset contains 83 missing values across three attributes.

Table 5: Missing values in the Employee dataset

Attribute	Missing values	Percentage
EnvironmentSatisfaction	25	0.57%
JobSatisfaction	20	0.45%
WorkLifeBalance	38	0.86%
Total	**83**	**1.88%**

1.6.3 Manager Dataset

The `manager` dataset contains three attributes: `EmployeeID`, `JobInvolvement`, and `PerformanceRating`. All 4,410 records are non-null, indicating that no missing values were detected.

1.6.4 Attendance Datasets

The `in_time` and `out_time` datasets each contain 262 attributes and 4,410 records. The available inspection output does not provide the non-null counts for individual columns. Therefore, their exact numbers of missing values need to be calculated separately before preprocessing.

1.7 Duplicate Records

Duplicate records were checked in all five datasets. The results showed that no duplicate records were detected in any of the files.

1.8 Initial Data Quality Assessment

Based on the initial inspection, the following observations were identified:

- All five datasets contain 4,410 records each, with a total of 555 attributes.
- The datasets contain numerical and string data types, requiring appropriate preprocessing techniques.
- The `general` dataset contains 28 missing values, while the `employee` dataset contains 83 missing values.
- The `manager` dataset contains no missing values.
- The exact number of missing values in the attendance datasets still needs to be determined.
- No duplicate records were detected in the five datasets.
- `EmployeeID` links all files (primary key in `general`, foreign key elsewhere), and the files can be read with UTF-8 encoding.

These findings provide an overview of the data structure and initial data quality. The next step is to examine the attendance records, handle missing values, and prepare suitable features for clustering.

2 Data Cleaning

The inspection in the previous section showed missing values in the `general`, `employee`, `in_time` and `out_time` files, no duplicate records, and attendance dates stored as strings. This section describes how each problem was handled to obtain the final table `df_clean`. The pipeline is implemented in the accompanying notebook.

2.1 Merging the Tables

The `general`, `employee` and `manager` files were joined on `EmployeeID` with `validate="one_to_one"`, which raises an error if any key is repeated. The merged table keeps all 4,410 employees and contains 28 columns. The two attendance files are processed separately (Section 2.4) and joined at the end.

2.2 Removing Duplicates

Duplicates were checked again after merging, because joins can silently multiply rows. Two criteria were used: fully identical rows and repeated `EmployeeID`. Both checks returned 0, so no record was removed and the table keeps 4,410 employees.

2.3 Handling Missing Values

Missing values were not treated in the same way everywhere. The decision depends on *why* a value is missing.

Table 6: Missing-value handling strategy

Dataset	Attribute	Missing	Strategy
General	NumCompaniesWorked	19	Median imputation
General	TotalWorkingYears	9	Median imputation
Employee	EnvironmentSatisfaction	25	Mode imputation
Employee	JobSatisfaction	20	Mode imputation
Employee	WorkLifeBalance	38	Mode imputation
Manager	—	0	No action
In/Out-time	all date columns	109,080 each	Not imputed: interpreted as absence and converted to features

2.3.1 General and Employee datasets

The General and Employee datasets contain 28 (0.63%) and 83 (1.88%) missing values, respectively. Because these counts are small, all rows were retained instead of being deleted, so that the other attributes of the affected employees are not lost. The missing values were imputed with a statistic chosen according to the nature of each variable.

Count variables: median. `NumCompaniesWorked` and `TotalWorkingYears` are non-negative counts. Figure 1 shows that both are right-skewed (skewness 1.12 and 1.03): most employees have low values, while a small number have very high ones (up to 40 years of experience and up to 9 previous companies) that form a long upper tail. These extreme values pull the mean above the median (11.28 vs. 10 years; 2.69 vs. 2 companies), so the mean no longer describes a typical employee, and for an integer count it also produces non-integer values. The median is not affected by the tail, so it was used.

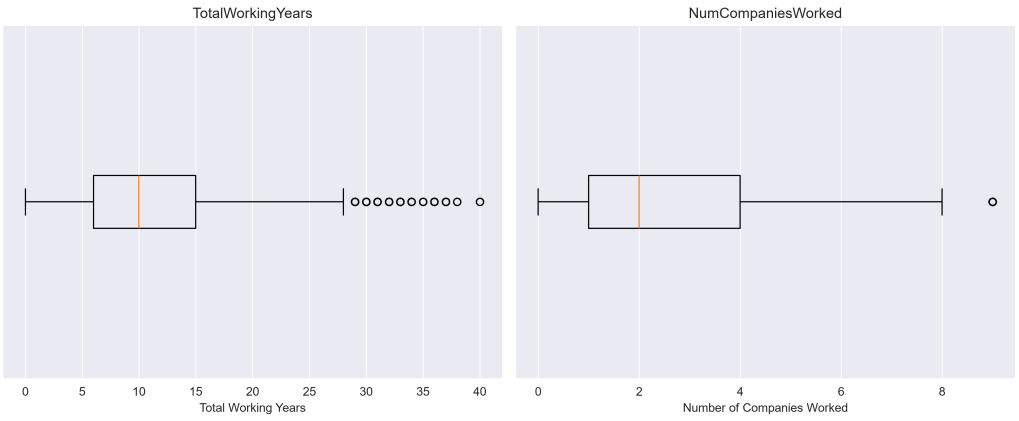


Figure 1: Distribution of `TotalWorkingYears` and `NumCompaniesWorked`, with mean (dashed) and median (solid).

Ordinal survey scores: mode. `EnvironmentSatisfaction`, `JobSatisfaction` and `WorkLifeBalance` are ordinal scores from 1 to 4. Figure 2 shows that their means (2.72, 2.73 and 2.76) fall between two levels and do not correspond to any valid answer. The mode (3, 4 and 3) is the level chosen most often, is always a valid score, and keeps the discrete nature of the variable, so it was used instead of the mean.

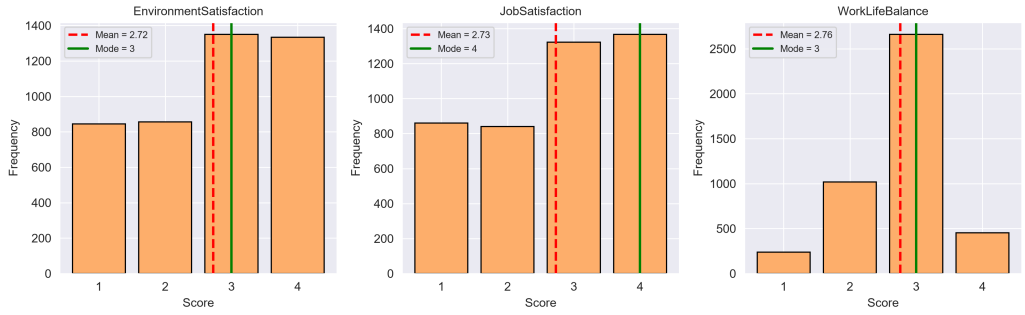


Figure 2: Distribution of the three survey scores, with mean (dashed) and mode (solid).

2.4 Processing Attendance Data

Each of `in_time` and `out_time` has 261 date columns and 109,080 missing cells. These are *not* random gaps. A cell is empty when the employee did not swipe on that day, that is, on a public holiday, on leave or when absent. Filling them with a mean or median would invent working hours that never happened. The cells were therefore handled in five steps:

Step 1. Convert to datetime. All 261 date columns were parsed with `pd.to_datetime`.

Step 2. Remove non-working days. Days on which no employee has a swipe-in and swipe-out are public holidays or closed days and carry no information. 12 such columns (the float columns in Table 3) were dropped. They account for $12 \times 4{,}410 = 52{,}920$ of the missing cells in each file, and the remaining $109{,}080 - 52{,}920 = 56{,}160$ cells are individual absences. The number of working days is therefore $N_{\text{work}} = 249$.

Step 3. Validate records. A day is valid only if both swipe-in and swipe-out exist and the duration lies in $(0, 24]$ hours. The check found 0 days with only one of the two swipes and 0 days with an invalid duration, so no record had to be discarded.

Step 4. Aggregate per employee. The remaining missing days are skipped (not filled) when computing the features below.

Step 5. Join. The three features are merged into the main table on `EmployeeID` (one-to-one).

$$h_{i,d} = t_{i,d}^{\text{out}} - t_{i,d}^{\text{in}} \quad \text{(hours, valid days only)}, \tag{1}$$

$$\text{AvgWorkHours}_i = \frac{1}{|D_i|} \sum_{d \in D_i} h_{i,d}, \tag{2}$$

$$\text{AbsentDays}_i = N_{\text{work}} - |D_i|, \qquad \text{AbsentRate}_i = \frac{\text{AbsentDays}_i}{N_{\text{work}}}, \tag{3}$$

where D_i is the set of valid days of employee i. Every employee has at least one valid day, so no median fallback was needed for these features.

Table 7: Summary of the engineered attendance features

Feature	**Mean**	**Std**	**Min**	**Median**	**Max**
AvgWorkHours	7.70	1.34	5.95	7.41	11.03
AbsentDays	12.74	5.50	1	13	24
AbsentRate	0.051	0.022	0.004	0.052	0.096

After this step the dataset contains **no missing values** (Table 9).

2.5 Standardizing Data Types

Table 8: Data type conversions

Attributes	**Before**	**After**	**Reason**
261 date columns of in/out-time	string	datetime	Needed to compute working hours
Survey scores, `NumCompaniesWorked`, `TotalWorkingYears`	float64	int64	Imputed values are whole numbers
`BusinessTravel`, `Department`, `EducationField`, `Gender`, `JobRole`, `MaritalStatus`	string	category	Memory and correct encoding
`Attrition`	string (Yes/No)	int64 (1/0)	External validation only

Three constant attributes (`EmployeeCount`, `Over18`, `StandardHours`) have a single value for all employees. They cannot separate clusters and were dropped.

A logical consistency check compared `YearsAtCompany` with `TotalWorkingYears`. Two employees (0.05%) have more years at the company than total working years. This is negligible and these records were kept unchanged.

2.6 Summary of Data Cleaning

Table 9: Dataset before and after cleaning

Criteria	Before	After
Employees (rows)	4,410	4,410
Missing values (general + survey)	111	0
Missing values (attendance, per file)	109,080	0 (converted to features)
Duplicate records	0	0
Constant attributes	3	0
Attributes (final table)	555 (5 files)	29

The 29 attributes of `df_clean` are: 21 attributes of `general` (24 minus 3 constant ones), 3 survey scores from `employee`, 2 ratings (`JobInvolvement`, `PerformanceRating`) from `manager`, and 3 attendance features. The cleaned table has one row per employee and is ready for exploratory data analysis and feature selection.

Listing 1: Core of the attendance cleaning step

```
in_dt  = in_df.apply(pd.to_datetime, errors="coerce")
out_dt = out_df.apply(pd.to_datetime, errors="coerce")

# drop public holidays: days with no swipe for every employee
holiday = in_dt.columns[in_dt.isna().all() & out_dt.isna().all()]
in_dt, out_dt = in_dt.drop(columns=holiday), out_dt.drop(columns=holiday
    )

# keep only days with both in and out
half = in_dt.isna() ^ out_dt.isna()
in_dt, out_dt = in_dt.mask(half), out_dt.mask(half)

hours = (out_dt - in_dt).apply(lambda s: s.dt.total_seconds() / 3600)
hours = hours.mask((hours <= 0) | (hours > 24))

att = pd.DataFrame({
    "AvgWorkHours": hours.mean(axis=1),          # NaN days skipped
    "AbsentDays":   hours.isna().sum(axis=1),
})
att["AbsentRate"] = att["AbsentDays"] / hours.shape[1]
```

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